Example 2.11

Which one of the points $i, -2+i$, and 3 is farthest from the origin?

Solution

The distance between origin to $z = i, -2+i,$ and 3 are

$$|z| = |i| = 1$$

$$|z| = |-2+i| = \sqrt{(-2)^2 + 1^2} = \sqrt{5}$$

$$|z| = |3| = 3$$

Since $1 < \sqrt{5} < 3$, the farthest point from the origin is 3.

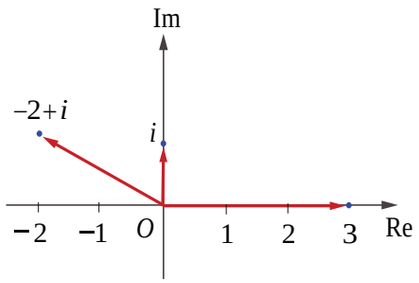


Fig. 2.19

Example 2.12

If z_1, z_2, and z_3 are complex numbers such that $|z_1| = |z_2| = |z_3| = |z_1 + z_2 + z_3| = 1$, find the value of $\left|\frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3}\right|$.

Solution

Since, $|z_1| = |z_2| = |z_3| = 1$,

$$|z_1|^2 = 1 \Rightarrow z_1\overline{z}_1 = 1, |z_2|^2 = 1 \Rightarrow z_2\overline{z}_2 = 1, \text{ and } |z_3|^3 = 1 \Rightarrow z_3\overline{z}_3 = 1$$

Therefore, $\overline{z}_1 = \frac{1}{z_1}, \overline{z}_2 = \frac{1}{z_2}$, and $\overline{z}_3 = \frac{1}{z_3}$ and hence

$$\left|\frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3}\right| = \left|\overline{z_1} + \overline{z_2} + \overline{z_3}\right|$$

$$= \overline{|z_1 + z_2 + z_3|} = |z_1 + z_2 + z_3| = 1.$$

Example 2.13

If $|z| = 2$ show that $3 \le |z + 3 + 4i| \le 7$

Solution

$$|z + 3 + 4i| \le |z| + |3 + 4i| = 2 + 5 = 7$$

$$|z + 3 + 4i| \le 7 \qquad (1)$$

$$|z + 3 + 4i| \ge \big||z| - |3 + 4i|\big| = |2 - 5| = 3$$

$$|z + 3 + 4i| \ge 3 \qquad (2)$$

From (1) and (2), we get $3 \le |z + 3 + 4i| \le 7$.

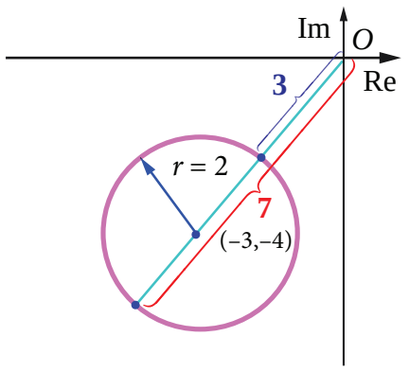


Fig. 2.20

Note

To find the lower bound and upper bound use $\big||z_1| - |z_2|\big| \le |z_1 + z_2| \le |z_1| + |z_2|$.

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